

WEEK - 7

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11.ReactJS-Handson

Initializing and creating npm-app

```
PS C:\Users\KIIT> npx create-react-app eventexamplesapp

Creating a new React app in C:\Users\KIIT\eventexamplesapp.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

added 1324 packages in 2m

269 packages are looking for funding
  run 'npm fund' for details

Initialized a git repository.

Installing template dependencies using npm...

added 17 packages, and changed 1 package in 8s

269 packages are looking for funding
  run 'npm fund' for details
Removing template package using npm...

removed 1 package, and audited 1341 packages in 5s

269 packages are looking for funding
  run 'npm fund' for details

9 vulnerabilities (3 moderate, 6 high)

To address all issues (including breaking changes), run:
  npm audit fix --force

Run 'npm audit' for details.

Created git commit.

Success! Created eventexamplesapp at C:\Users\KIIT\eventexamplesapp
Inside that directory, you can run several commands:

  npm start
    Starts the development server.

  npm run build
    Bundles the app into static files for production.

  npm test
    Starts the test runner.

  npm run eject
    Removes this tool and copies build dependencies, configuration files
    and scripts into the app directory. If you do this, you can't go back!

We suggest that you begin by typing:

  cd eventexamplesapp
  npm start

Happy hacking!
PS C:\Users\KIIT> cd eventexamplesapp
PS C:\Users\KIIT\eventexamplesapp> code .
```

Code in src/Components/Counter.js

```
src > Components > JS Counter.js > ...
1  import React, { Component } from "react";
2
3  class Counter extends Component {
4    constructor(props) {
5      super(props);
6      this.state = {
7        count: 0,
8      };
9    }
10
11    increment = () => {
12      this.setState({ count: this.state.count + 1 });
13    };
14
15    sayHello = () => {
16      console.log("Hello! Have a nice day 😊");
17    };
18
19    handleIncrement = () => {
20      this.increment();
21      this.sayHello();
22    };
23
24    decrement = () => {
25      this.setState({ count: this.state.count - 1 });
26    };
27
28    render() {
29      return (
30        <div>
31          <h2>Counter Value: {this.state.count}</h2>
32          <button onClick={this.handleIncrement}>Increment</button>
33          <button onClick={this.decrement}>Decrement</button>
34        </div>
35      );
36    }
37  }
38
39  export default Counter;
```

Code in src/Components/WelcomeButton.js

```
src > Components > JS WelcomeButton.js > ...
1  import React from "react";
2
3  function WelcomeButton() {
4    const sayWelcome = (message) => {
5      alert("Message: " + message);
6    };
7
8    return (
9      <div>
10        <button onClick={() => sayWelcome("Welcome to React Events Lab!")}>
11          Say Welcome
12        </button>
13      </div>
14    );
15  }
16
17  export default WelcomeButton;
18
```

Code in src/SyntheticClick.js

```
src > Components > JS SyntheticClick.js > ...
1  import React from "react";
2
3  function SyntheticClick() {
4    const handleClick = (e) => {
5      // e is a SyntheticEvent
6      alert("I was clicked!");
7      console.log("Synthetic Event Type: ", e.type);
8    };
9
10    return (
11      <div>
12        <button onClick={handleClick}>Click Me (Synthetic Event)</button>
13      </div>
14    );
15  }
16
17  export default SyntheticClick;
```

Code in src/Components/CurrencyConverter.js

```
src > Components > JS CurrencyConverter.js > ...
1  import React, { useState } from "react";
2
3  function CurrencyConverter() {
4    const [rupees, setRupees] = useState("");
5    const [euros, setEuros] = useState(null);
6    const conversionRate = 0.011; // Example: 1 INR = 0.011 EUR
7
8    const handleSubmit = (e) => {
9      e.preventDefault(); // Prevent page reload
10     const result = parseFloat(rupees) * conversionRate;
11     setEuros(result.toFixed(2));
12   };
13
14   return (
15     <div>
16       <h2>Currency Converter (INR ➡ EUR)</h2>
17       <form onSubmit={handleSubmit}>
18         <input
19           type="number"
20           value={rupees}
21           onChange={(e) => setRupees(e.target.value)}
22           placeholder="Enter INR"
23         />
24         <button type="submit">Convert</button>
25       </form>
26       {euros !== null && <p>Converted Value: € {euros}</p>}
27     </div>
28   );
29 }
30
31 export default CurrencyConverter;
```

Code in src/App.js

```
src > JS App.js > default
1  import React from "react";
2  import Counter from "../Components/Counter";
3  import WelcomeButton from "../Components/WelcomeButton";
4  import SyntheticClick from "../Components/SyntheticClick";
5  import CurrencyConverter from "../Components/CurrencyConverter";
6
7  function App() {
8    return (
9      <div className="App" style={{ padding: "20px", fontFamily: "Arial" }}>
10        <h1>🔥 React Event Examples App</h1>
11        <Counter />
12        <hr />
13        <WelcomeButton />
14        <hr />
15        <SyntheticClick />
16        <hr />
17        <CurrencyConverter />
18      </div>
19    );
20  }
21
22  export default App;
```

Output:

 React Event Examples App

Counter Value: 0

Increment Decrement

Say Welcome

Click Me (Synthetic Event)

Currency Converter (INR  EUR)

Enter INR

The image displays three sequential screenshots of a web application titled "React Event Examples App".

Top Screenshot: The application shows a "Counter Value: 0" and a "Currency Converter (INR ₹ EUR)" section. A dark modal dialog is open, displaying the message "Message: Welcome to React Events Lab!" with an "OK" button. The browser's address bar shows "10.35.89.115:3000".

Middle Screenshot: The application is in the same state, but the modal dialog now displays "I was clicked!".

Bottom Screenshot: The application shows the "Counter Value: 0" and "Currency Converter (INR ₹ EUR)" section. The "Converted Value: € 13.20" is displayed below the input field. The browser's address bar shows "1200".